

# Project Documentation

Master overview: architecture, tech stack, feature summary, and documentation index.

|         |                                                                               |
|---------|-------------------------------------------------------------------------------|
| Program | Diploma in Advanced AI & Software Engineering                                 |
| Batch   | 25/26                                                                         |
| Project | Smart Hire — Service Marketplace                                              |
| Team    | Nilan Iddawela, Janaka Eranda, Dasun Anjana, Damith Lasantha, Matheesha Milan |
| Date    | 21 July 2026                                                                  |

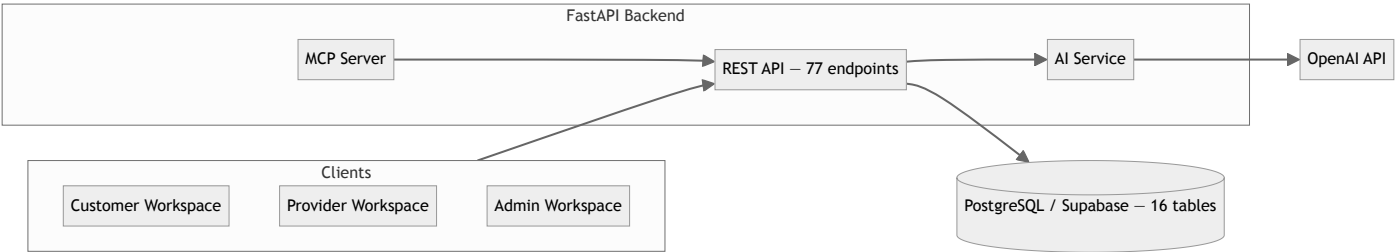
## 1. Project Overview

**Smart Hire** is a centralized digital marketplace connecting customers with skilled service providers — plumbing, electrical work, home maintenance, tutoring, cleaning, repairs, and technical support. Customers discover, compare, and book verified providers; providers publish services and manage their own bookings; an integrated AI assistant helps with recommendations and can act on a customer's behalf once they confirm.

## 2. Technology Stack

| LAYER           | TECHNOLOGY                                                                                   |
|-----------------|----------------------------------------------------------------------------------------------|
| Frontend        | Angular (standalone components, signal-based state, no zone.js)                              |
| Backend         | Python, FastAPI, SQLAlchemy ORM                                                              |
| Database        | PostgreSQL, hosted on Supabase                                                               |
| AI              | OpenAI Chat Completions API, native tool-calling; standalone MCP server for external clients |
| Auth            | JWT bearer tokens, bcrypt password hashing                                                   |
| Testing         | pytest (backend), Vitest via Angular's builder (frontend)                                    |
| Version Control | Git, hosted on GitHub                                                                        |

## 3. System Architecture



## 4. Feature Summary by Role

| ROLE     | KEY FEATURES                                                                                                                                                    |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Customer | Browse/search/filter services, favourites, booking requests, cancellations, payments, reviews, direct messaging, floating AI chat, full AI assistant page       |
| Provider | Publish services, weekly availability, coverage areas, accept/reject/complete bookings, messaging, AI review summaries                                          |
| Admin    | User status/verification management, category management, service moderation, review moderation, payment management, platform dashboard with real revenue chart |

## 5. Project Documentation Index

This project's documentation set, all in `academic_documentation/` :

| DOCUMENT                                           | COVERS                                               |
|----------------------------------------------------|------------------------------------------------------|
| Smart Hire SRS.pdf                                 | Functional and non-functional requirements           |
| Smart_Hire_Use_Case_Diagrams_Documentation.pdf     | Actors and use case diagrams per role                |
| SmartHire_User_Stories_Documentation.pdf           | User stories grouped by project epic                 |
| SmartHire WBS.pdf                                  | Full task breakdown and delivery timeline            |
| SmartHire Database Design.pdf                      | ER diagram, table dictionary, integrity rules        |
| <b>Smart Hire Database Scripts.pdf</b>             | Real, runnable schema DDL                            |
| SmartHire_API_Documentation.pdf                    | Full REST API reference                              |
| SmartHire_Testing_documentation.pdf                | Test strategy, coverage, defects found and fixed     |
| <b>Smart Hire Deployment Guide.pdf</b>             | How to set up and run the system                     |
| <b>Smart Hire Source Code &amp; Repository.pdf</b> | Repository structure, branches, contribution history |
| Smart Hire Lessons Learned.pdf                     | 24 real development hiccups and fixes                |
| Smart Hire — Jira Project Management Summary.pdf   | Project management via Jira, board summary           |

## 6. Team

| MEMBER          | PRIMARY CONTRIBUTION AREA                                                 |
|-----------------|---------------------------------------------------------------------------|
| Nilan Iddawela  | Backend APIs, admin, AI/MCP, messaging, QA, git workflow, documentation   |
| Janaka Eranda   | Authentication & account status, database schema, profile/dashboard pages |
| Dasun Anjana    | Bookings, notifications, availability (frontend & backend)                |
| Damith Lasantha | AI feature backend & recommendation logic                                 |
| Matheesha Milan | Landing page, services/categories/search backend                          |